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Optimizing HPC Performance for CDC-Based Storage and Reconstruction of High-Resolution Imagery

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IDRT

Introduction

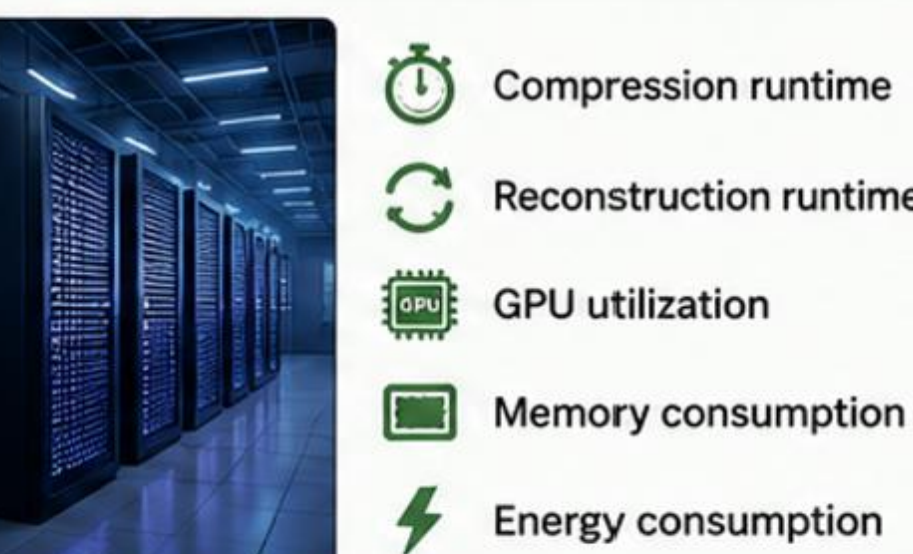
Advances in drone technology have enabled the collection of ultra-high-resolution imagery, but a single mission can generate hundreds of gigabytes of data, creating challenges for storage, transfer, and processing. This study evaluates a drone imagery compression workflow on HPC systems by examining how compression ratios and reconstruction tile sizes affect runtime, throughput, GPU utilization, memory usage, and image quality. We also assess the impact of these settings on downstream object detection performance. The goal is to identify configurations that maximize storage savings and computational efficiency while maintaining image quality and computer vision performance.

STORAGE COST (LONG-TERM)



Higher Cost as Data Volumes Grow

COMPUTATIONAL COST



Higher Cost as Compression Increases



RESEARCH QUESTION

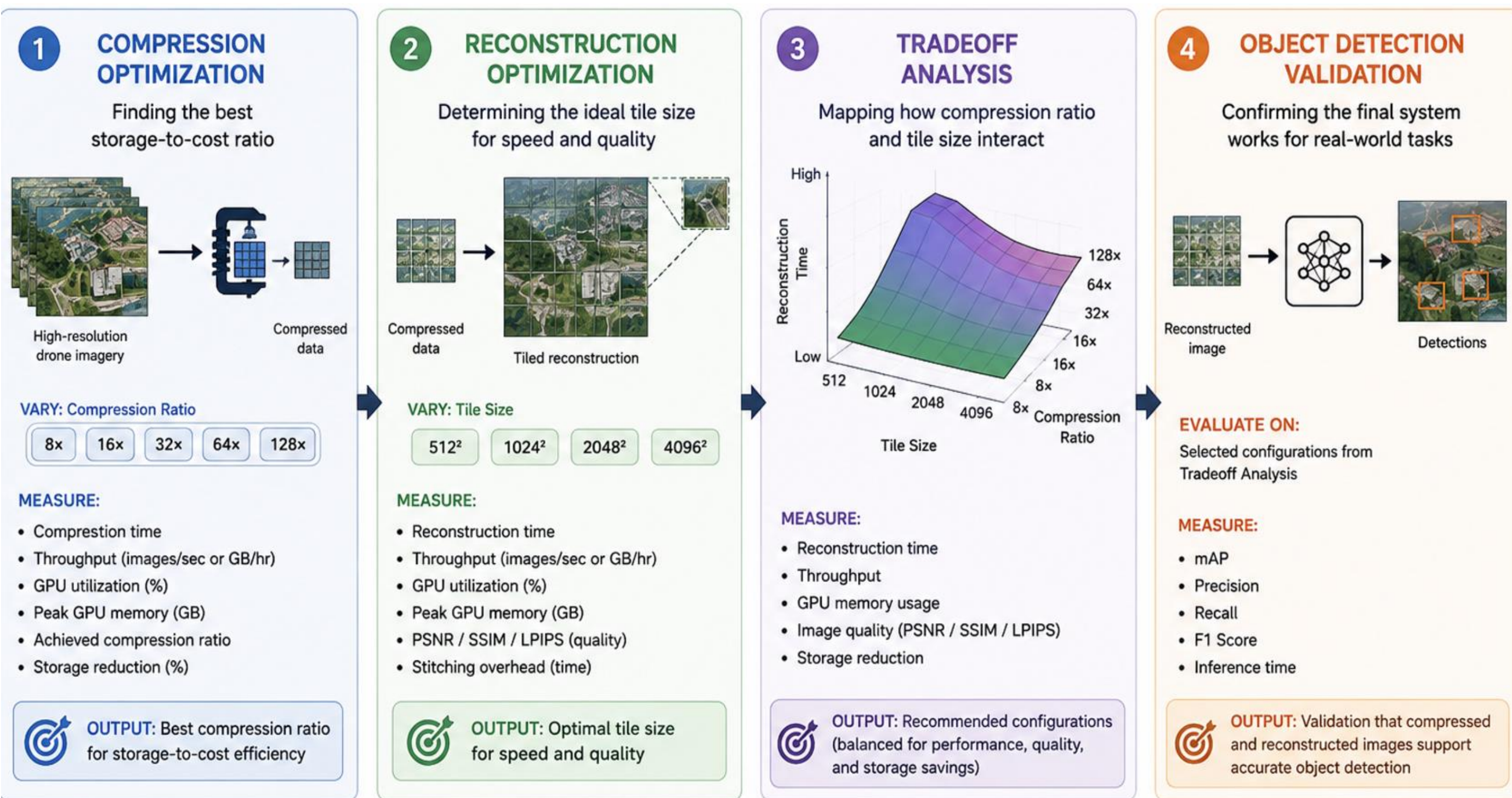
Can HPC-enabled compression reduce long-term storage requirements enough to justify the computational cost of compression and reconstruction?

Data Description

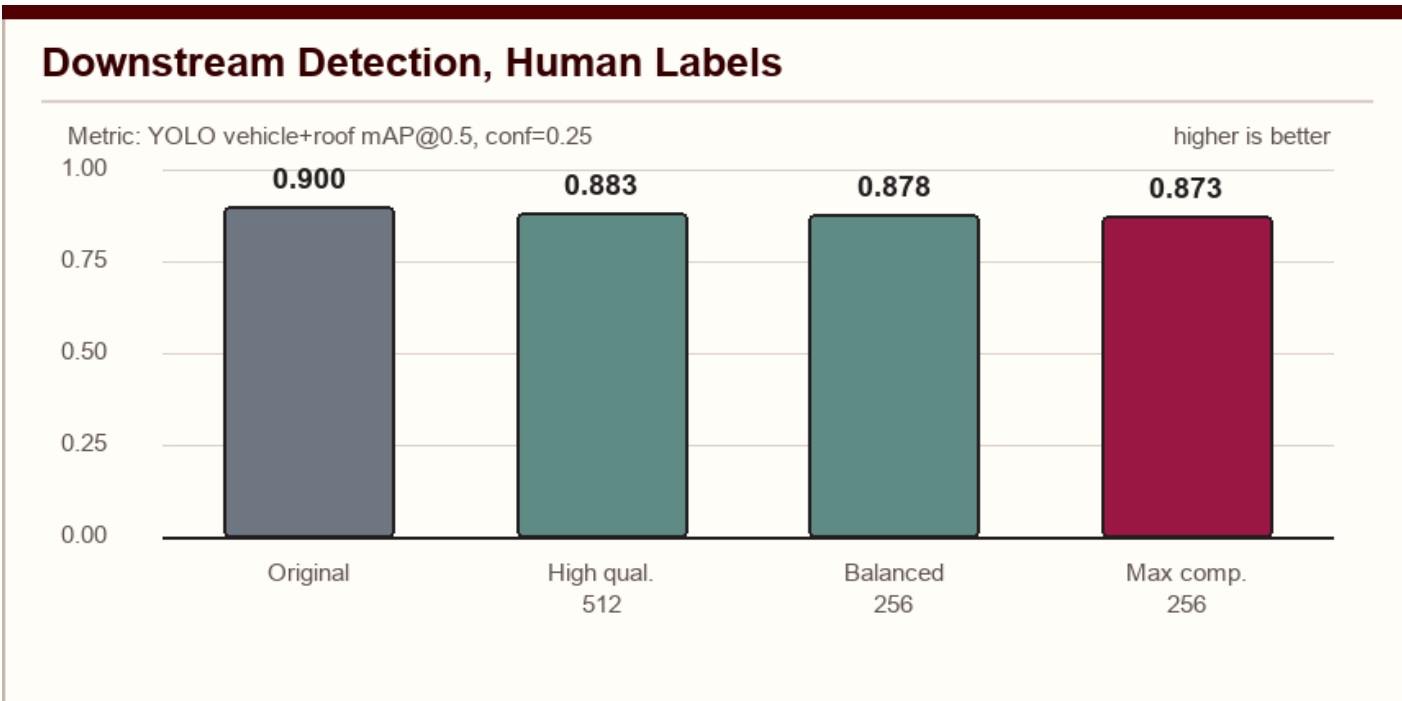
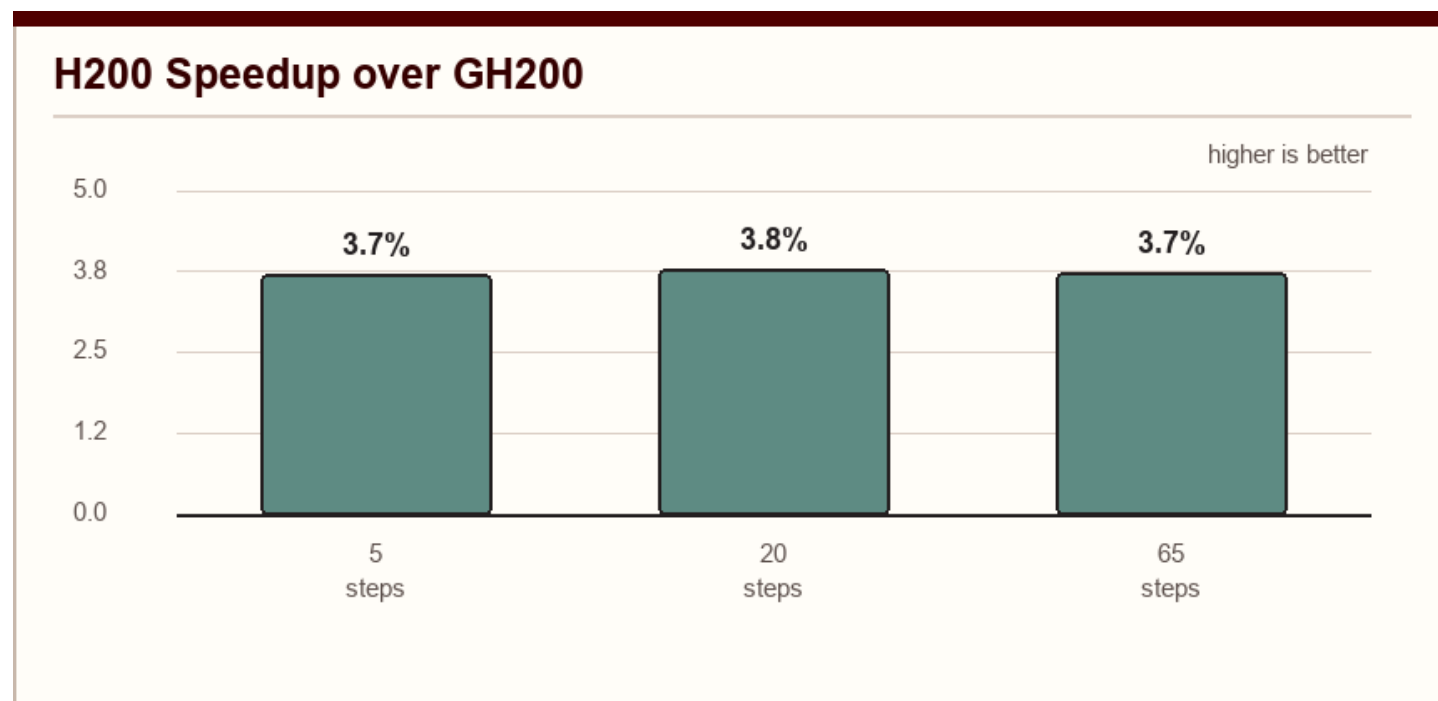
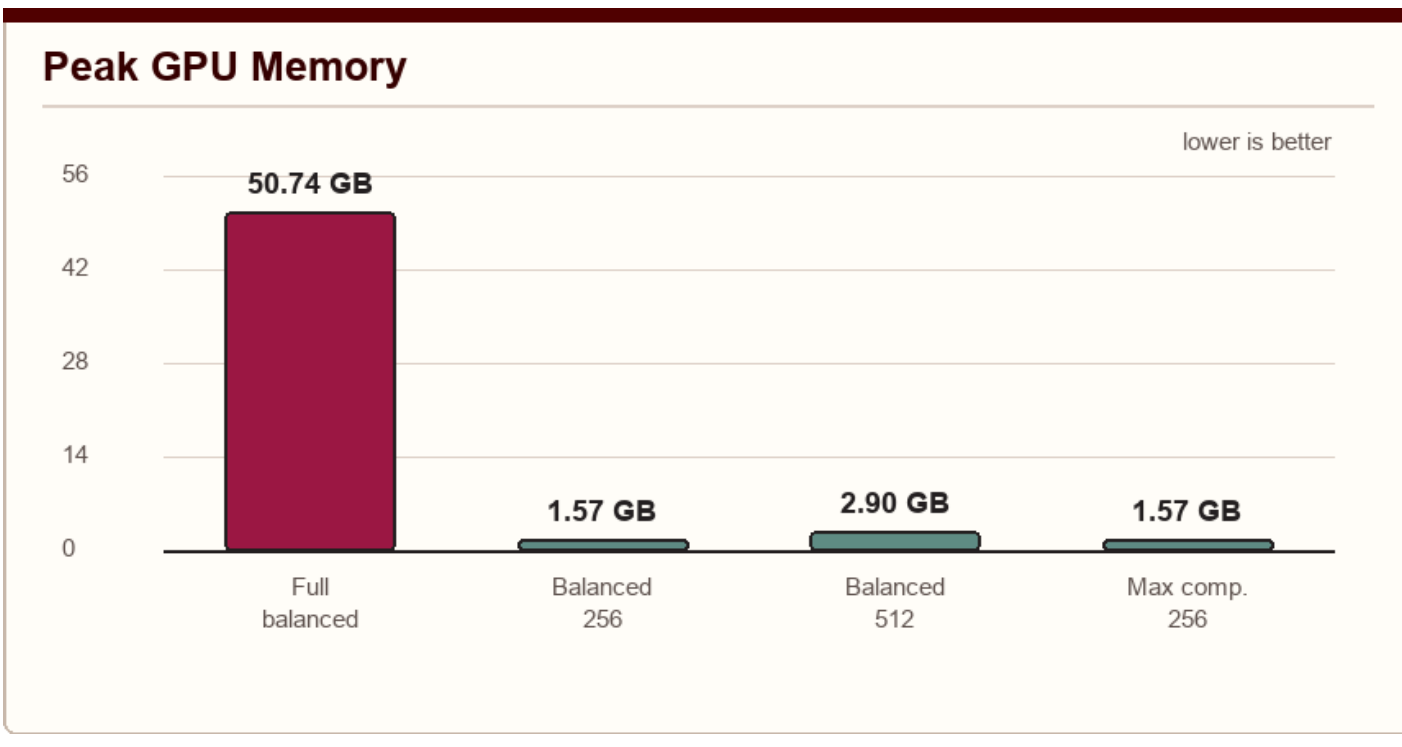
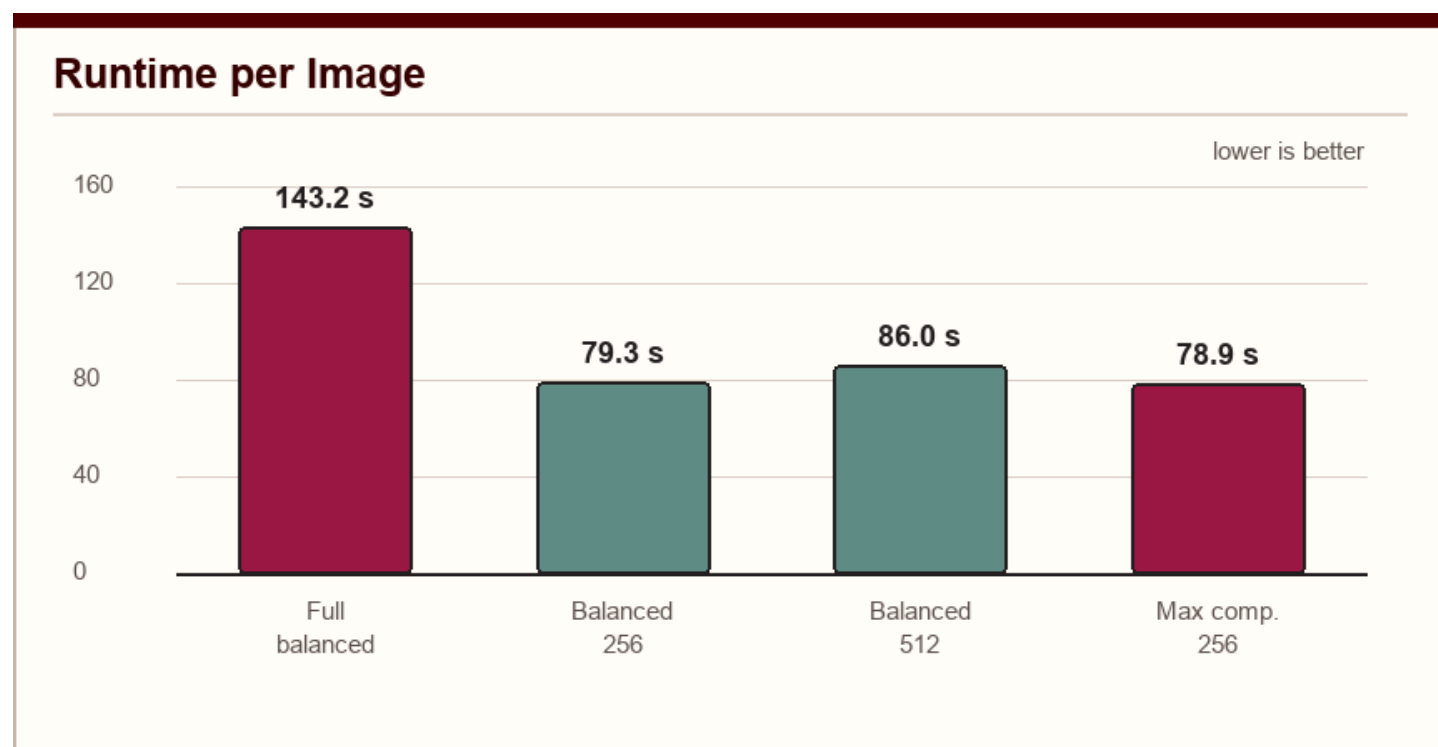
- Study area: Galveston, Texas
- 100 RGB drone images acquired at altitude of approx. 47 m
- Average original size of 8.34 MB, 24-bit color depth
- Spatial resolution: 5472 X 3648 pixels



Experiment Design



Resource Efficiency and Bottleneck Summary



Measured Proxies, Not Hardware Telemetry				
	Full balanced	Balanced 256	Balanced 512	Max comp. 256
Runtime pressure	100%	55%	60%	55%
GPU memory pressure	50.7 GB	1.6 GB	2.9 GB	1.6 GB
Non-GPU overhead	0.2%	5.9%	5.0%	6.0%
Quality-loss proxy	33.29 dB	33.14 dB	33.23 dB	30.89 dB
Detection retention	0.900	0.878	0.883	0.873
SAM roof IoU	1.000	0.899	0.900	0.896

Results

- Recommended setting:** balanced checkpoint b00064 with 256 x 256 tiling. 512 x 512 is the quality-safe backup.
- Tiling wins on cost:** 256 x 256 tiling cuts peak GPU memory from ~52 GB to ~1.6 GB and runtime from ~143 s to ~79 s per image.
- Detection holds up:** YOLOv8s loses only ~1.4% F1 at balanced_256; SAM masks usable on all 50 images.

Reconstruction tradeoff matrix (N50, LPIPS)

Config	Sec/img	GPU GB	Ratio	PSNR	SSIM
High quality 512	88.5	2.96	29.7x	34.19	0.938
Balanced 256 ★	79.3	1.57	80.0x	33.14	0.877
Balanced 512	86.0	2.90	78.5x	33.23	0.878
Max compression 256	78.9	1.57	139.9x	30.89	0.819

Object detection impact — YOLOv8s vehicle + roof (N50)

Image set	mAP@.5	mAP@.5:.95	Precision	Recall	F1
Original	0.864	0.607	0.884	0.882	0.883
High quality 512	0.845	0.488	0.893	0.869	0.880
Balanced 256 ★	0.839	0.481	0.877	0.864	0.870
Max compression 256	0.833	0.475	0.879	0.857	0.868

Baseline compression matrix (full-image)

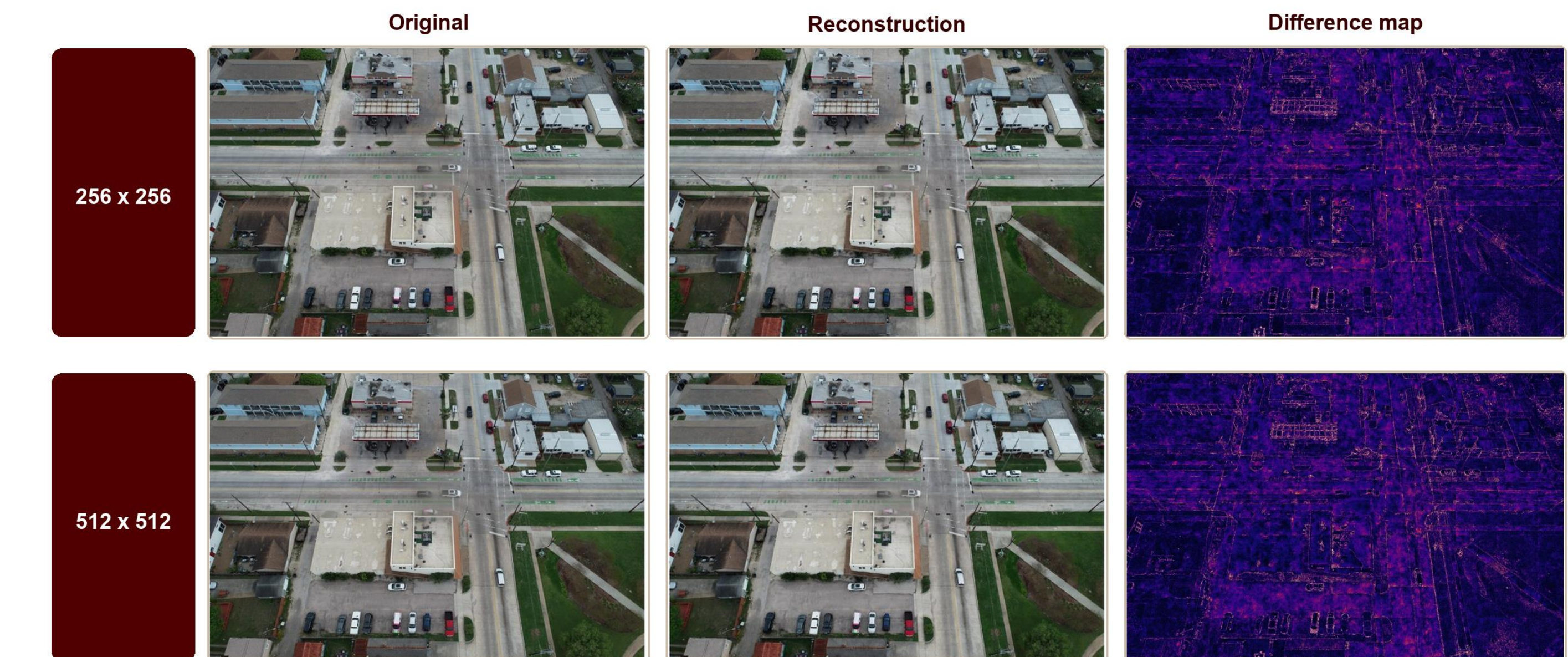
Role	Ckpt	Ratio	PSNR	SSIM
High quality	b00512	32.67x	34.84	0.9423
Balanced	b00064	93.11x	33.73	0.8803
High compression	b00128	165.73x	31.52	0.8281
Baseline	b02048	71.20x	29.45	0.8727

SAM mask stability (vehicle / roof prompts)

Prompt	Config	IoU	Dice	Fail%
Vehicle	HighQual 512	0.735	0.838	0%
Vehicle	Balanced 256	0.735	0.838	0%
Vehicle	MaxComp 256	0.733	0.837	0%
Roof	HighQual 512	0.900	0.945	0%
Roof	Balanced 256	0.899	0.945	0%
Roof	MaxComp 256	0.896	0.943	0%

Visual Reconstruction Check: 256 vs 512 Tiling

Representative Galveston drone scene, only image-level comparison panels are shown for poster readability.



Columns compare the same scene: original image, reconstructed image, and absolute reconstruction-error map.

Qualitative panel - 256 x 256 vs 512 x 512 tiling

Discussion

- Tiling, not downscaling, is the right lever: full detail at ~1.6 GB peak.
- Detection is robust: F1 within ~2% of original, so balanced_256 is safe.
- mAP@.5:.95 is more sensitive (~21% drop); prefer 512 for fine localization.
- YOLOv8s beats YOLO-World; SAM masks stay stable.

Conclusion / Future Direction

- Full-image CDC is feasible but memory-heavy; 256 tiling is the recommended operating point.
- Report CDC/PCDC as a joint systems + image-quality problem.
- Next: expand validation and visual panels.